DISPATCH
@IETF127

LinkID: location-independent references

The LinkID URI Scheme and Resolution Model (draft-linkgenetic-linkid-uri-01)

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Motivation

- **References outlive their locations.** Court decisions, legislation and technical documents cite URLs that fail in two ways:
  - **Link rot** (visible): the URL no longer resolves. 38% of web pages from 2013 were gone a decade later · Pew 2024
  - **Content drift** (silent): the URL still works, the content has changed. Affected 3 of 4 checked web references in scholarly articles · Jones et al. 2016; 13% of working links in NYT articles · Zittrain et al. 2021
  - Together: ~50% of URLs in U.S. Supreme Court opinions no longer led to the cited content · Zittrain et al. 2014
- **Existing PIDs persist the identifier, not the target.** 4–15% of sampled DOIs end in an error · Klein, Balakireva 2020
- **Goal:** an interoperable, vendor-independent way to resolve a stable reference to its current target, detect target changes, with defined lifecycle states and verifiable answers.
- **Interest beyond Link Genetic:** discussions with Wikimedia Foundation (T422252), Internet Archive and DataCite. No third-party implementation commitment yet.

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Background (1/2): draft and prior discussion

- **Draft:** draft-linkgenetic-linkid-uri-01: linkid:<uuid>, resolution model, HTTPS binding, lifecycle states (active, revoked, tombstoned …)
- **IANA:** provisional registration of linkid:
- **Prior discussion:** uri-review (T. Hardie); W3C TPAC 2025 breakout, Kobe; WICG proposal #238
- **DISPATCH list, Oct 2026:** reviews by E. Rescorla (motivation vs. DOI, scale, security) and J. Klensin (why not a URN?)
- **IPR (patent pending):** disclosure #7635; royalty-free commitment for conforming implementations: PATENTS.md
- **Existing WGs:** no prior engagement

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Background (2/2): running code and list feedback

- **Implementation:** resolver at linkid.io; Ed25519-signed, offline-verifiable portable resolution records (separate from platform ECDSA evidence); open-source failover resolver prepared on Wikimedia Toolforge, not yet in service (whitepaper); SDKs JS/Python/Java, Apache-2.0: github.com/Link-Genetic-Inc/lid
- **Feedback and direction for -02:**
  - DOI exists → DOI persists the identifier, not the target; a DOI is a valid LinkID target
  - Scale, trust in the resolver → signed records; mirrors verify and cache
  - Takeover of identifiers → per-authority keys and signed transfers (today: one platform key, audited updates)
  - Why not a URN? → the substance is resolution and lifecycle; open to a URN namespace
- **Still open:** operator discovery without a central list, key recovery; governance model proposed, not yet constituted (backup)

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Desired dispatch outcome

Requested outcome: additional discussion and community development, e.g. a new non-WG mailing list on persistent reference resolution.

Questions for the community:

- Is resolver-independent resolution with signed records in scope for the IETF?
- A dedicated linkid: scheme, or a URN namespace for the same resolution model?
- Once the design settles: is AD sponsorship a realistic path?

Links:

- Draft: datatracker.ietf.org/doc/draft-linkgenetic-linkid-uri
- Spec and SDKs: github.com/Link-Genetic-Inc/lid
- Resolver: linkid.io · Slides and FAQ: linkid.linkgenetic.com/ietf-127
- Contact: c.n@linkgenetic.com

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LinkID

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